Telescoping sums

High school · sequences · Progressive exercises · 10

Exercise 1. Find $\sum[\frac{1}{k} - \frac{1}{k+1}]$ from $k = 3$ to 10.

Exercise 2. Express $\frac{1}{k(k+2)}$ as a difference of two fractions.

Exercise 3. Find $\sum(2k + 1)$ from $k = 0$ to n using $(k + 1)^2 - k^2$.

Exercise 4. If $S_n = \frac{n}{n+1}$, find the least integer $n \geq 1$ with $1 - S_n < 0.01$.